

* Timers: programming with timers:

* Q6. Programming with timers:

Develop a 'C' code and necessary settings to toggle on board LED with a time period of 500ms using Timer-2 and assume clock frequency = 16 MHz.

program:

```
#include "main.h"
```

```
/* private Variable */
```

```
TIM_HandlertypeDef htim2;
```

```
int main(void)
```

/* Initialize all configured peripherals */

```

Mx_GPIO_Init();

```

```

Mx_TIM2_Init();

```

```
HAL_TIM_Base_Start_IT(&htim2);
```

while(1)

 Σ

void HAL_TIM_PeriodElapsedCallback(TIM_HandleTypeDef * htim)

HAL_GPIO_TogglePin (GPIOD, GPIO_PIN_15);

2

* STM32CubeMx configuration:

* STM32CubeMX Configuration:

1) System Core

- RCC → High speed clock (HSE) → crystal/ceramic Resonator
- SYS → Debug → Serial wire

sys $\rightarrow$ Debug $\rightarrow$ Serial wire

2) Timers → TIM2 (Timer 2) → clock source Internal clock
parameter settings:

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```
prescaler: 1000
```

Counter Mode: up

counter period (Auto Reload): 8000

auto-reload preload: **Enable**

3) NVIC settings → NVIC Interrupt Table Enabled
TIM2 Global Interrupt ☒

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* Timer calculation: Number to be loaded for the given prescale value (PSC) and T_{out} .

$$T_{out} = \frac{(PSC \times \text{preload})}{\text{clock frequency}}$$

$$500 \times 10^{-3} = \frac{(1000 \times \text{preload})}{16000000}$$

$$0.5 = \frac{\text{preload}}{16000}$$

$$\boxed{\text{preload} = 8000}$$

* Counters and Timers:

↳ Timer: The clock comes from an Internal source such as PLL, XTAL and RC

↳ Counter: The clock comes from an external source and fed to the processor pin. It is called as event-counter since it counts the event happening outside the CPU.

↳ The STM32Fxx ARM comes with a large number of timers. The timers are called TMx where $x = 0, 1, 2, 3, 4$ and so on.

↳ Categories of Timers:

1) Advanced control timer: used for PWM and DC motor control applications.

2) General purpose timer: Generic timer/counter applications.

Timer type	Timer	Resolution	type	prescaler factor	channels
Advanced control	TIM1, TIM8	16-bit	up, down, up/down	Any Integer b/n 1 and 65536	4
General purpose	TIM2, TIM5	32-bit	up, down, up/down	- 11 -	4
	TIM3, TIM4	16-bit	up, down, up/down	- 11 -	4
	TIM9	16-bit	up	- 11 -	2
	TIM10, TIM11	16-bit	up	- 11 -	1
	TIM12	16-bit	up	- 11 -	2
	TIM13 TIM14	16-bit	up	- 11 -	1
	TIM6, TIM7	16-bit	up	- 11 -	0
Basic					